

MORINGA DSF-FT12



Outline

- ❖ Introduction
- ❖ Objectives
- ❖ Data Understanding
- ❖ Data Cleaning
- ❖ Data Analysis
- ❖ Statistical Analysis
- ❖ Recommendations



01.

Introduction

- SilverScreen Studios is preparing to enter the movie production industry and seeks a data-driven approach to guide its entry strategy.
- This project analyzes historical movie industry data to uncover the drivers of box office success and return on investment.

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02.

Objectives

- Identify the top-performing directors and genres, and evaluate their impact on the movie market.
- Analyze the most successful studios and the revenue they generate.
- Determine the relationship between production budget and movie revenue
- Investigate whether the release month significantly affects movie revenue.

03.

Data Understanding

- The datasets used in this analysis are compiled from reputable online movie databases
- They cover various dimensions of the movie industry e.g genres, gross revenue, production budgets, directors



04.

Data Cleaning

- Dropped missing and irrelevant data.
- Standardized formats and corrected inconsistencies for consistency.
- Filtered variables to focus on factors relevant to movie market analysis

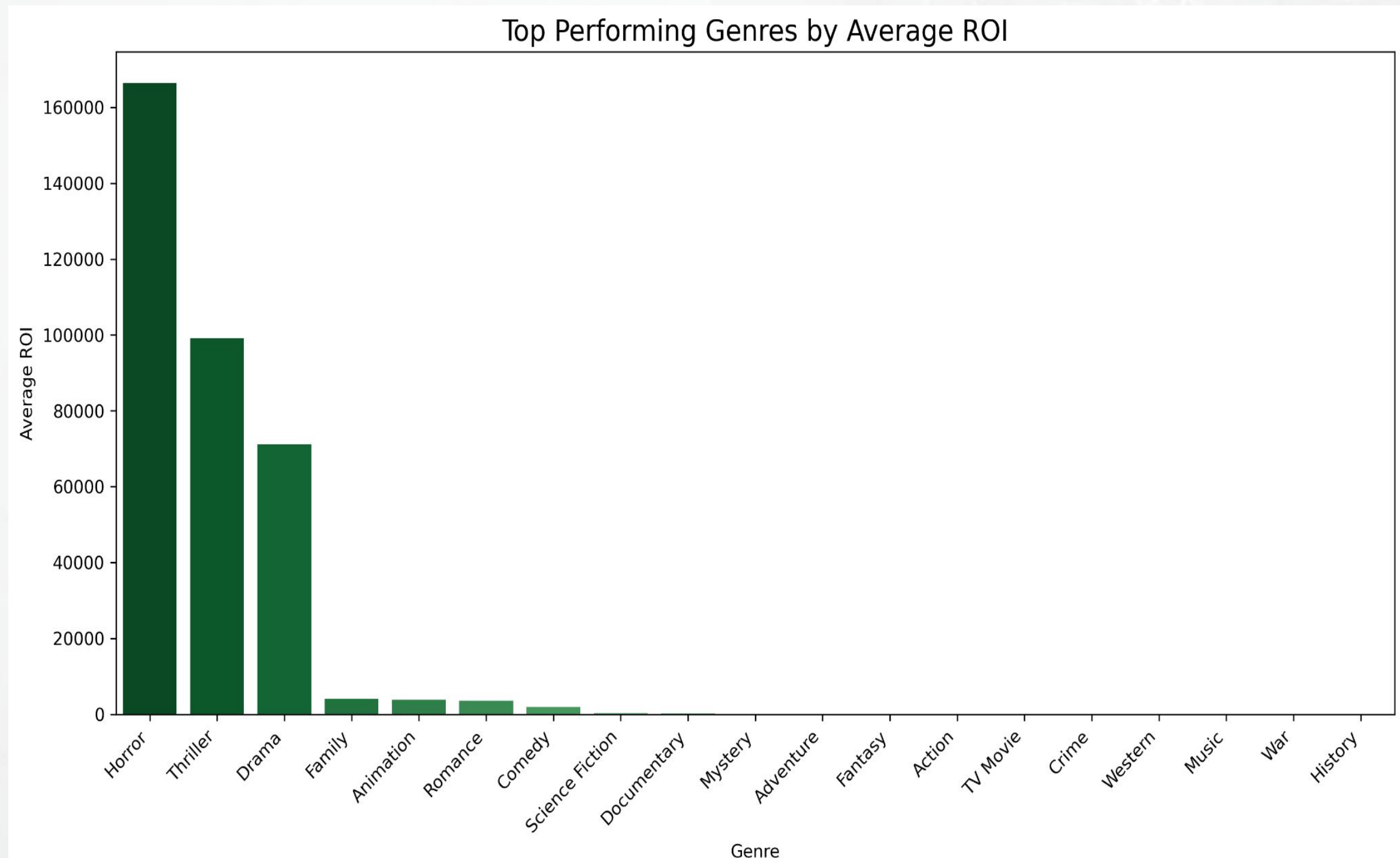


05.

Data Analysis

Which genres have the highest Return On Investment(ROI)?

- Horror and Thriller genres are the top performers in terms of ROI.
- The company should focus on genres with a high ROI.
- This shows that the genres deliver more return per dollar invested, even if their total revenue is not always the highest.



05.

Data Analysis

Who are the top performing directors we can learn from and hire for production?

Key Insights from Movie Data (86,000+ Titles)

- Top directors drive popularity
- Identified using a hybrid score based on ratings and vote counts.
- Top directors consistently produce popular films that attract large audiences.

Top 10 Directors Ranked by Hybrid Score

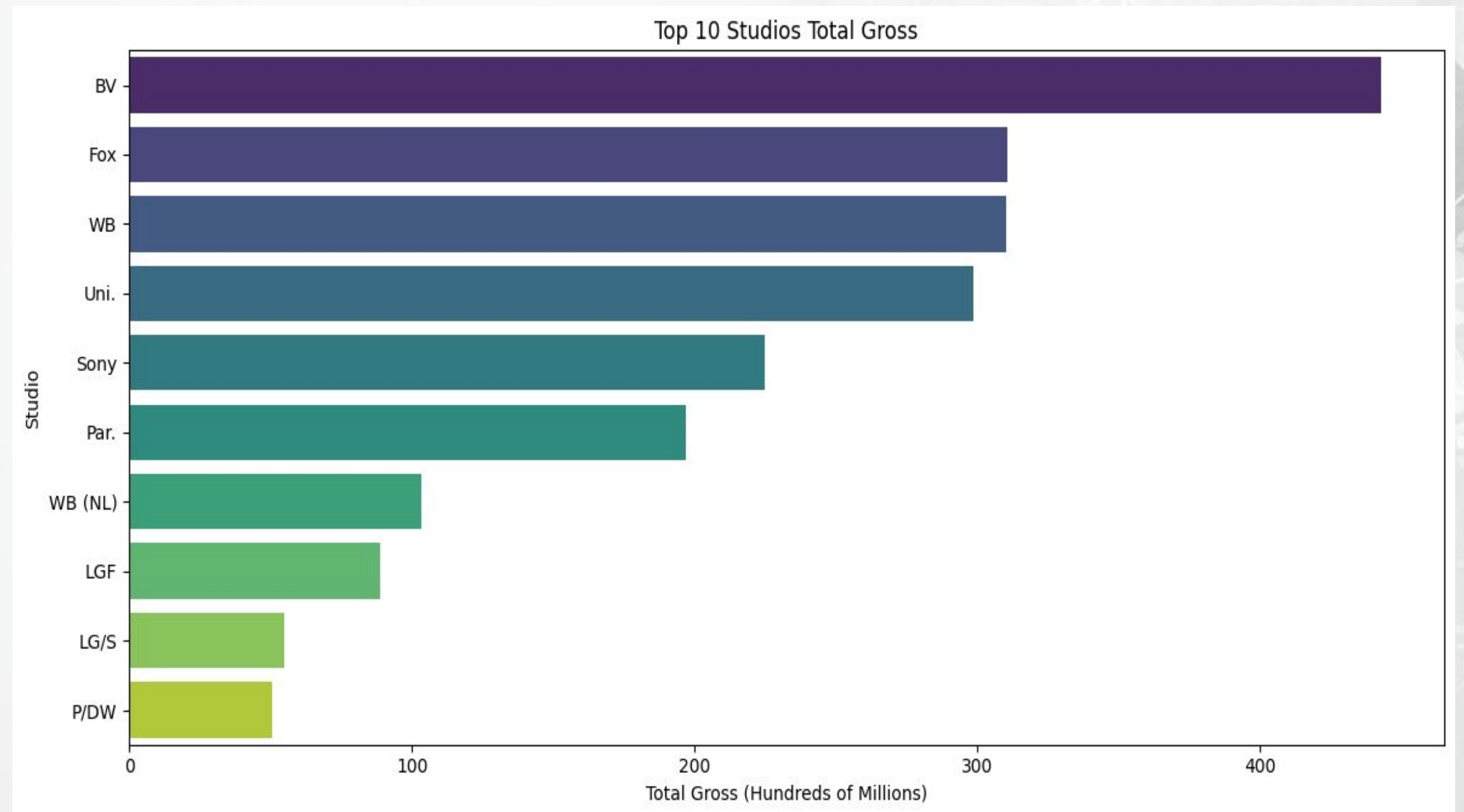
	director_name	avg_rating	total_votes	num_movies	weighted_rating	consistency	score
0	Christopher Nolan	8.43	4,994,749	4	8.33	2.52	7.46
1	Wes Anderson	7.93	1,033,599	3	7.61	6.14	7.39
2	Quentin Tarantino	8.68	1,655,377	4	8.38	1.24	7.31
3	Derek Cianfrance	7.30	443,101	3	6.88	9.09	7.21
4	David Fincher	7.87	1,717,750	3	7.67	4.58	7.20
5	Anthony Russo	8.23	2,361,820	4	8.05	1.94	7.13
6	Joe Russo	8.23	2,361,820	4	8.05	1.94	7.13
7	Justin Lin	7.17	886,483	3	6.94	7.97	7.10
8	Ryan Coogler	7.47	809,006	3	7.17	6.14	7.01
9	Taika Waititi	7.75	713,498	4	7.36	4.96	7.00

05.

Data Analysis

Which studios make more money ?

- Data from over 3000 movies show the top 3 as Buena Vista Studios, Fox Studios and Warner Bros.
- The top studios rake in hundreds of millions, therefore their actions should be modelled.

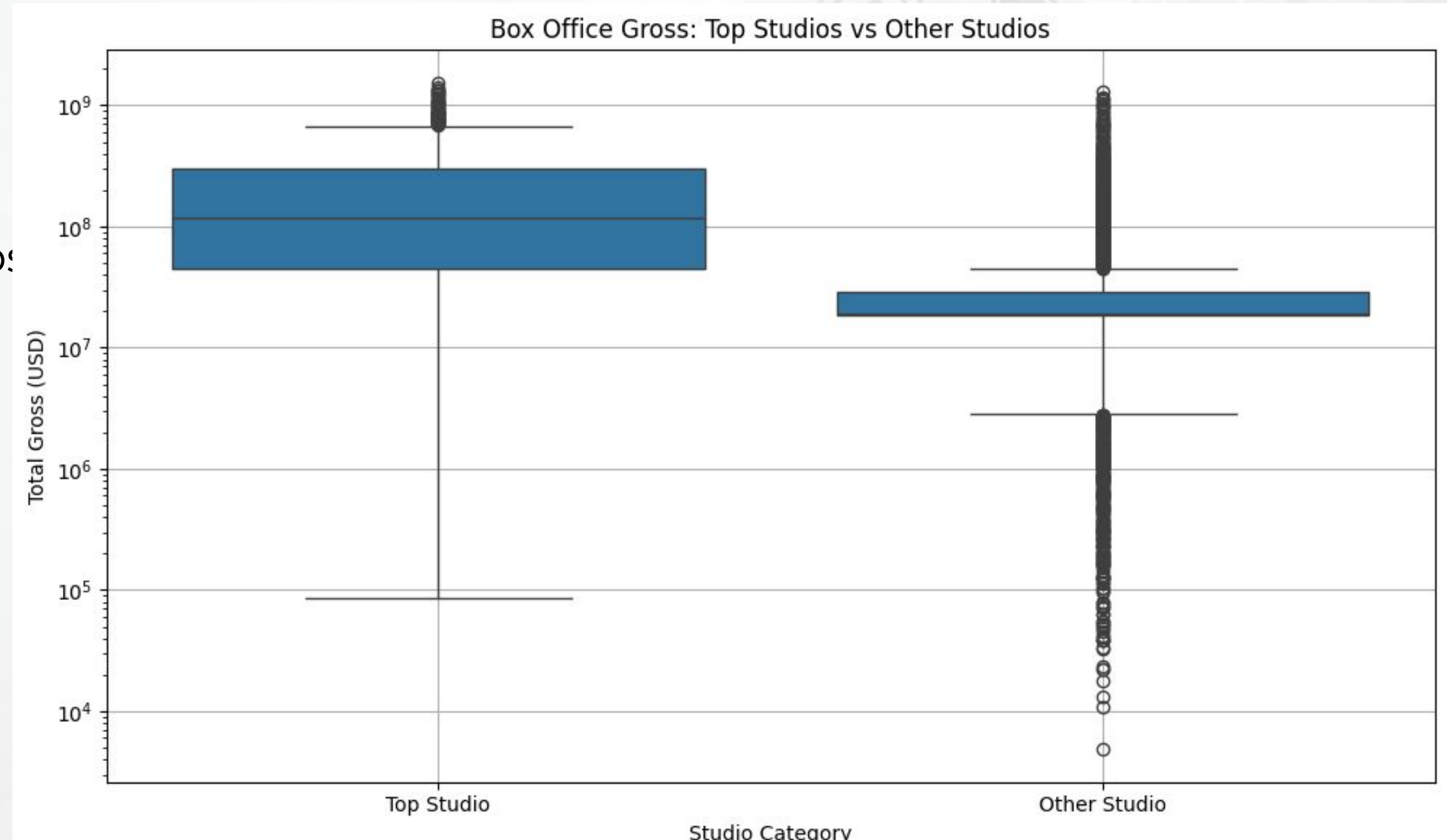


05.

Data Analysis

Are there differences between the top studios and others?

- Analysis of top studios and other studios show average total revenue generated by movies from the top 10 studios is NOT the same.
- There are real, underlying differences in performance between these studios.

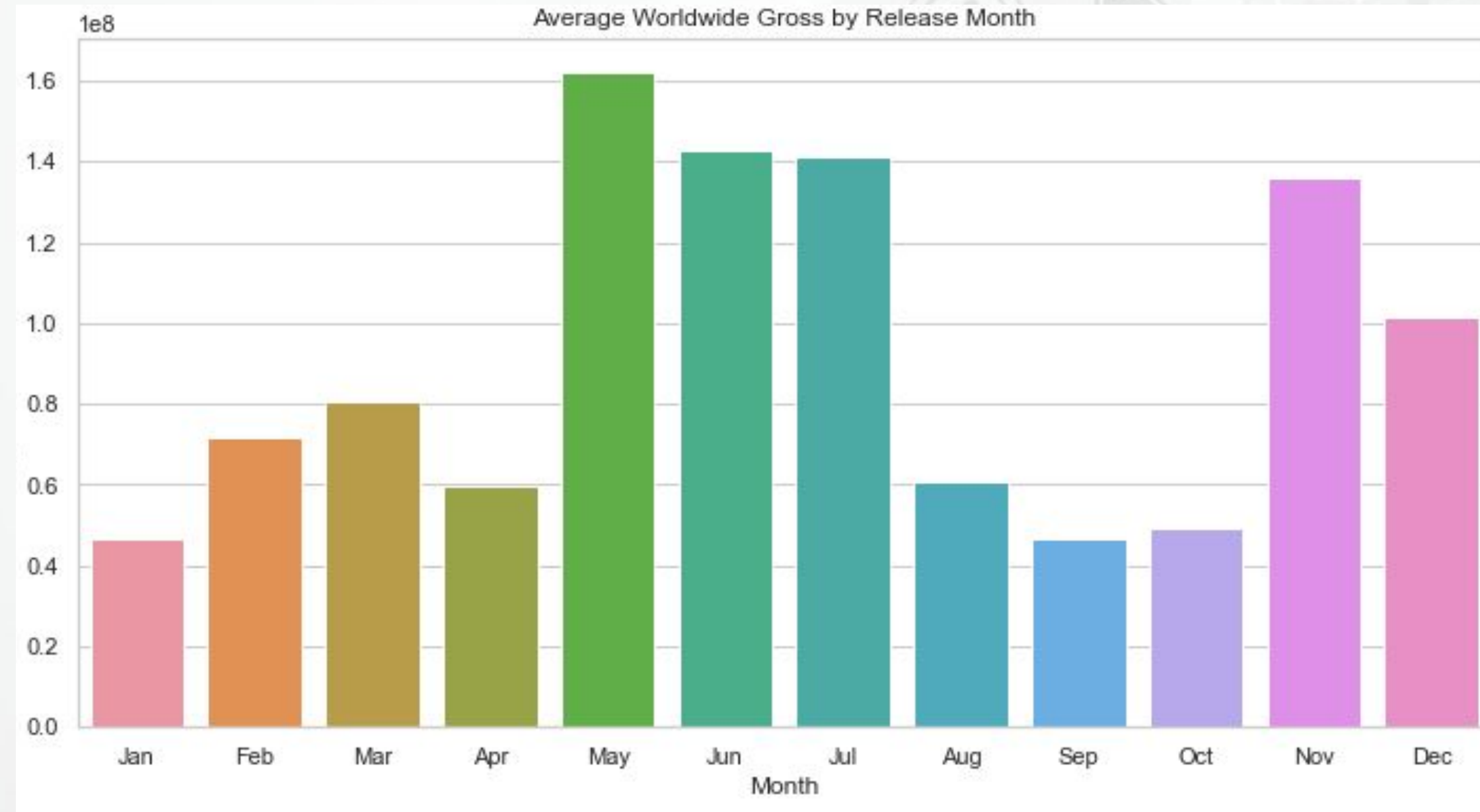


05.

Data Analysis

When is the best time to release a movie

- Timing matters. Strategic release scheduling, especially in summer and late-year holidays, can significantly enhance a movie's box office performance.

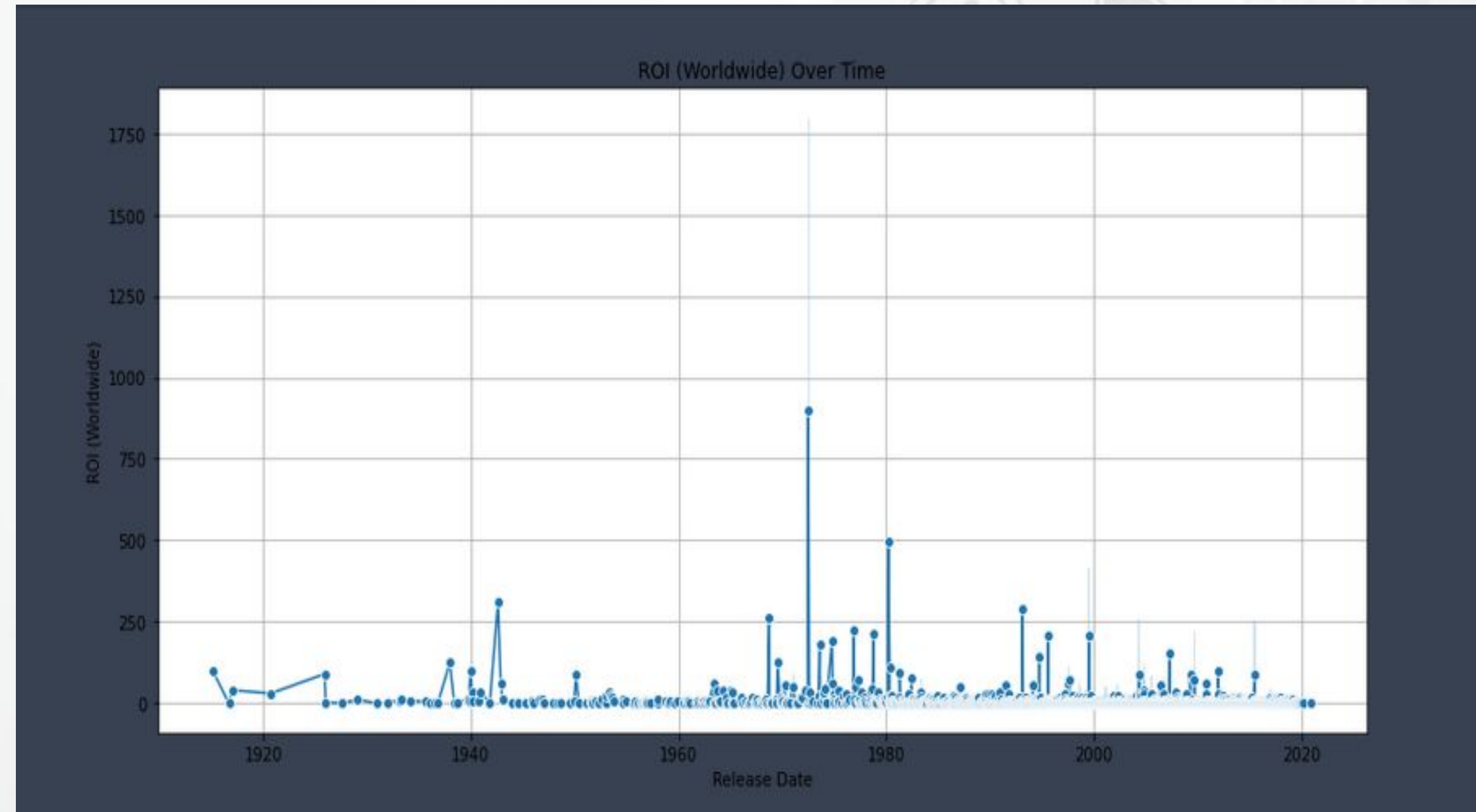


05.

Data Analysis

ROI Trends Over The Years(Worldwide)

- ROI fluctuates significantly year to year
- No consistent upward or downward trend, indicating ROI is heavily influenced by external market factors and blockbuster hits.
- ROI spiked in the late 2000s–early 2010s, likely from global box office growth

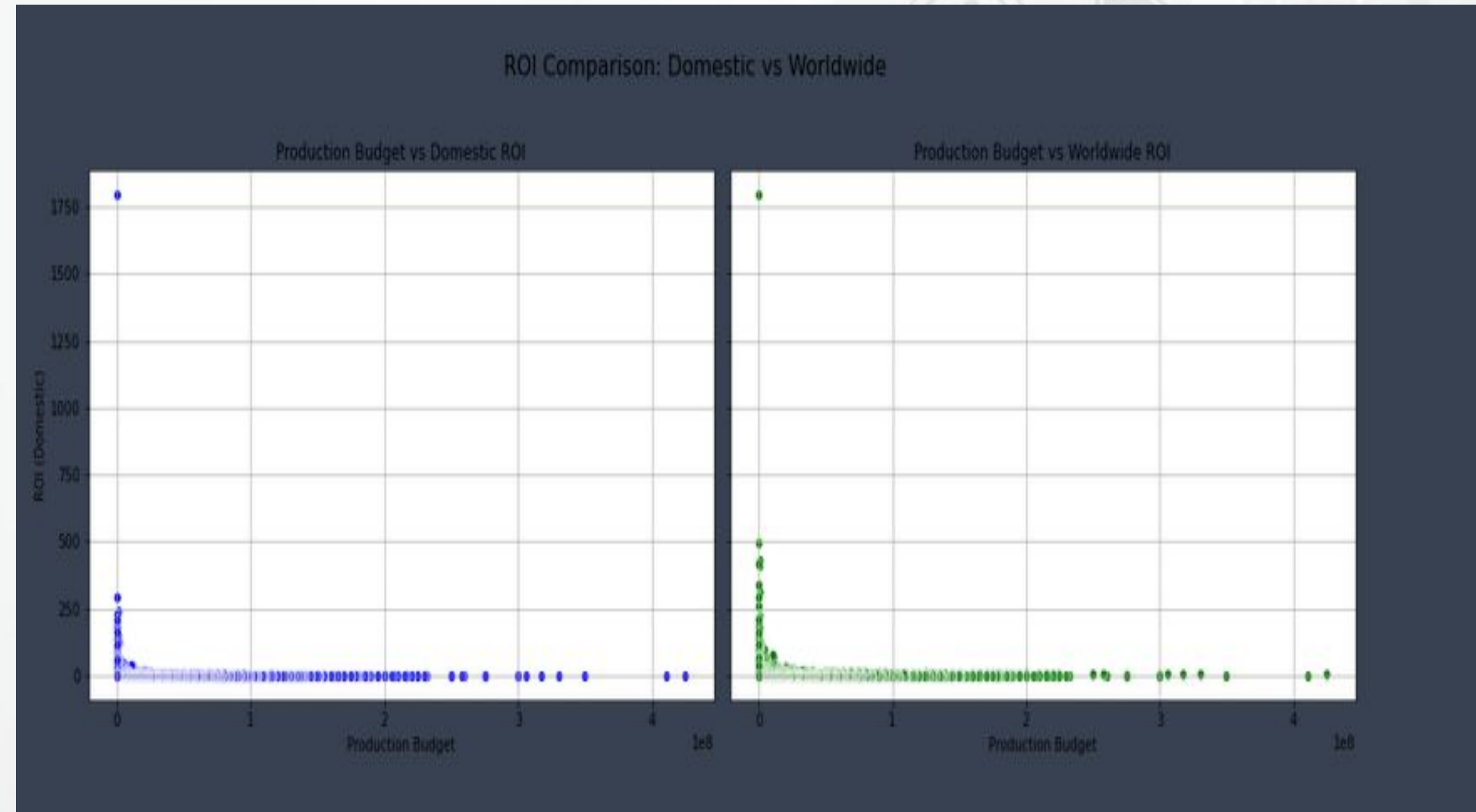


05.

Data Analysis

Production Budget vs ROI (Domestic & Worldwide)

- Most low-budget films have low or negative ROI
- Worldwide ROI generally trends higher than domestic ROI for the same production budget.
- No strong linear correlation, suggesting budget alone doesn't guarantee profitability.



06.

Statistical Analysis

Anova Test

- (H_0): The mean worldwide gross revenue is the same across all release months.
- (H_1): At least one release month has a significantly different mean revenue.
- Results - F-statistic: 12.209, p-value: 0.000
- The p-value indicates strong evidence against the null hypothesis. Therefore, it can be concluded that average worldwide gross revenue differs significantly between at least two release months.

06.

Statistical Analysis

Independent T test

- (H_0): There is no difference in average worldwide gross between May and November releases.
- (H_1): There is a difference in average worldwide gross between May and November releases.
- Results - T-statistic: -1.07, P-value: 0.2833
- The p-value is greater than the commonly used significance level of 0.05, so the null hypothesis cannot be rejected. This indicates that there is no statistically significant difference in average worldwide gross revenue between movies released in May and those released in November

06.

Recommendations

- Focus on genres with a high ROI (Horror and Thriller)
- Partner with top directors to boost movie visibility and revenue by increasing audience reach
- For higher revenues movies should be released around summer or around late holidays
- Tighten budget control, especially in mid-tier productions. Avoid over-investment in genres where high budgets do not translate to proportional profits.

Meet the team

- ❖ Kelvin Kipkorir
- ❖ Raphael Muthenya
- ❖ Sharon Aoko
- ❖ Lucy Mutua
- ❖ Charles Mutembei
- ❖ Andrew Chege



Q&A

Thanks

